

GS EXTERIOR EXPERTS · DECK DIVISION · PERMIT SET

20 x 20 Deck · Castle Rock · Permit Set

20' x 20' · 10' high · ledgered · south Castle Rock · 2x10 SYP @ 12" on two
4x12 DF beams, 8 Diamond Piers · TimberTech Vintage Coastline PVC, picture
frame, no butt joints · Fulton black + drink rail · Hardie skirt · stair with
landing · October 02, 2026

SHEET INDEX

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Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. This set carries no pricing.

20 x 20 deck · G-001 · General notes · design criteria

G-001GENERAL NOTES · DESIGN CRITERIA		20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...	
Loads, codes, structure summary, sheet index, site — the permit-set cover		GSX · Jade Pender 303-550-9558 · October 02, 2026 · Rev 0 · PERMIT SET	
PROJECT		SHEET INDEX	
20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle Rock		G-001	General notes · design criteria
Castle Rock CO 80104		A-001	Isometric · renders
396 SF · 10'-0" above grade · 1 zone		S-100	Foundation plan
DESIGN CRITERIA		S-101	Framing plan
Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf	A-101	Decking plan
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs	A-201	Railing plan + elevation
Dead load	10 psf deck	A-301	Section + front elevation
Roof snow (cover)	—	D-501	Details
Design total	50 psf on the deck	SITE	
Wind	115 mph Vult, Exposure C, Risk Cat. II (ASCE 7-16) — confirm special wind region	Setbacks not on file — verify the plat / ILC before layout	
Frost depth	3'-0" (footings bear below)		
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report		
Seismic	B		
Wildfire (WUI)	not in a WUI zone		
Codes	2021 IRC as adopted by Castle Rock; IRC R507 decks; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9		
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs		
STRUCTURE			
Footings	DP-50/50		
Posts	12 x 6x6		
Beams	DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb); DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 21 2x10 #1 True Frame Joist SYP GC @ 12" OC x 19'-5"		
Joists	2-ply 2x10 front and sides		
Rims	2x10 ledger x 19'-9 7/16" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners		
Ledger	TimberTech Vintage Coastline — 41 rows parallel to the house, picture frame in Vintage Coastline, 3/16" gaps, Cortex		
Decking	Fulton 36" Black: 8 bays, 10 posts, 54 LF + drink rail		
Rail			
No scale		Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.	

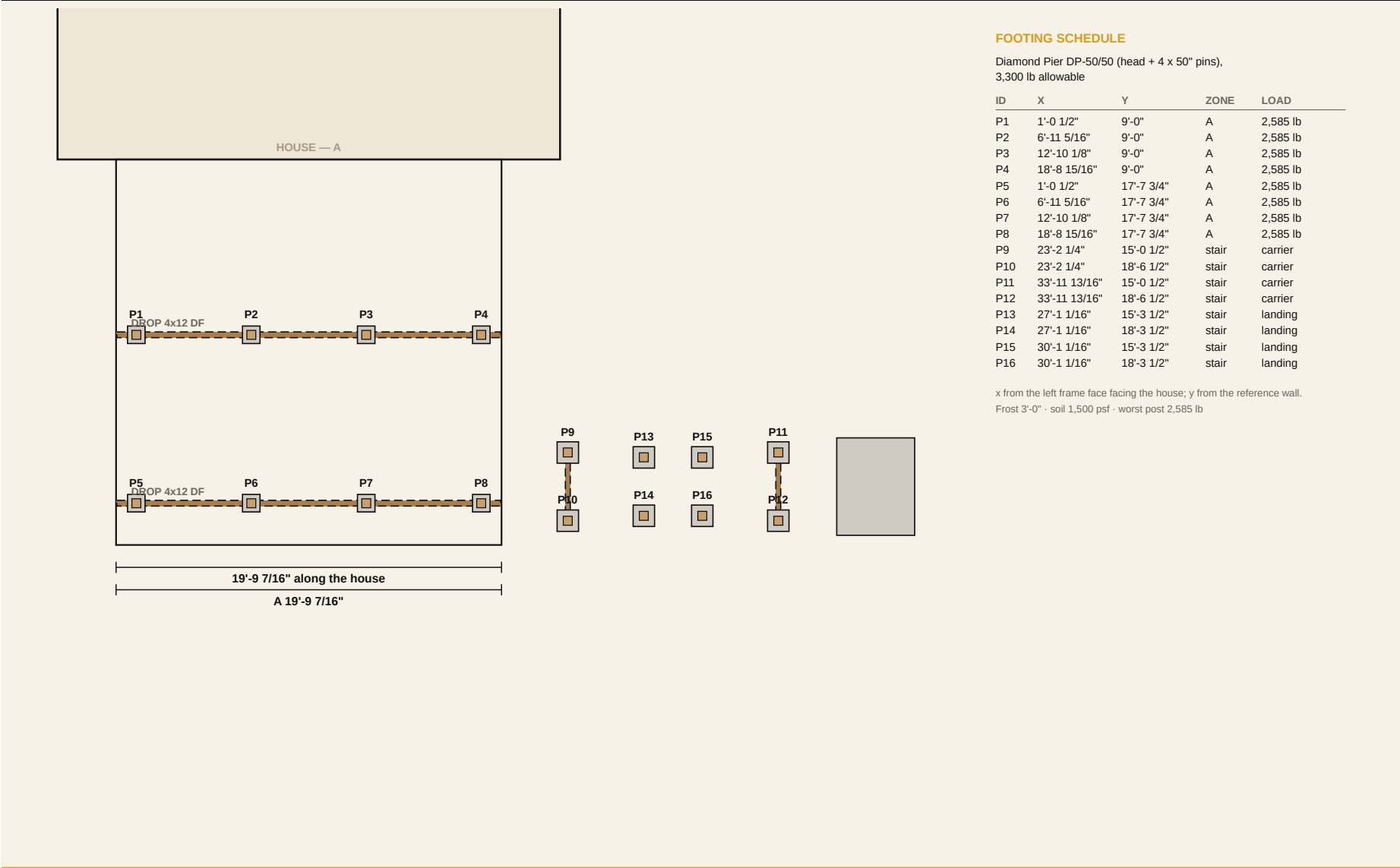
20 x 20 deck · S-100 · Foundation plan

S-100

FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...
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20 x 20 deck · S-101 · Framing plan

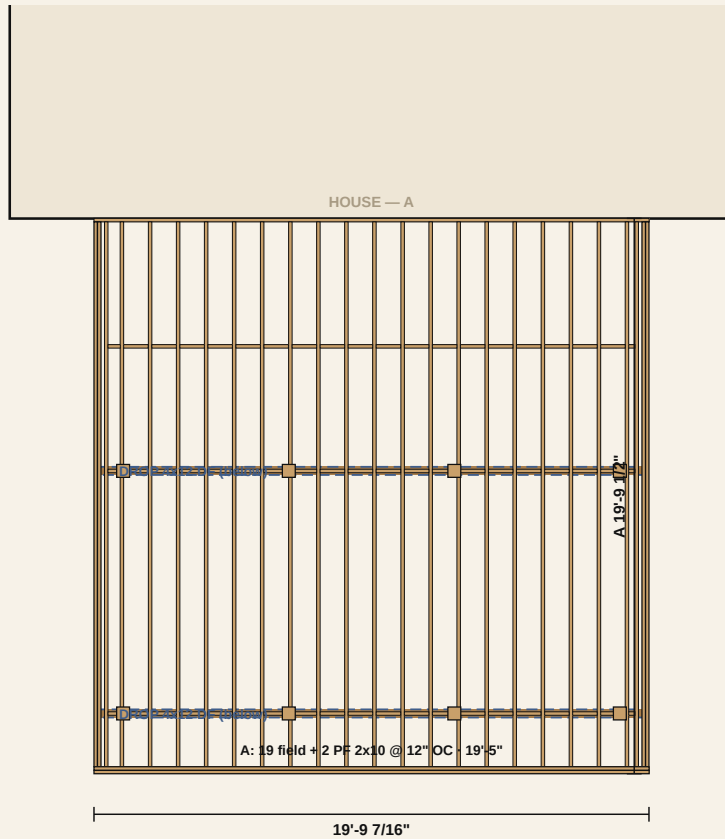
S-101

FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...

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FRAMING NOTES

21 2x10 #1 True Frame Joist SYP GC @ 12"
OC x 19'-5"

2-ply 2x10 front and sides

2x10 ledger x 19'-9 7/16" total ·
LedgerLOK 2 rows 2" from top/bottom,
staggered 12" OC, first 3" from each end
blocking one row over the beam,
staggered

G-Tape 2" every joist, PF joist,
blocking and the ledger top · 4" on both
rims and the beam(s)

DROP 4x12 DF: 19'-8 7/16", 4 posts at
1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" /
18'-8 15/16" (worst 2,585 lb); DROP 4x12

Connectors:

Simpson LUS28Z face-mount hanger
2x8/2x10: 42

Simpson HUCQ210-2-SDS concealed-flange
double 2x10 hanger (SDS screws
included): 4

Simpson DTT1Z lateral tie kit w/ SDS
screws: 4

Simpson H2.5AZ hurricane tie (joists to
the drop beam, hidden): 46

Simpson BC46Z post cap 6x post / 4x
beam: 8

7/16" x 4-1/2" HDG hex bolt + nut + 2
washers: 44

Simpson LSCZ adjustable stair-stringer
connector: 10

Plan · not to scale · house at the top · dimensions govern

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20 x 20 deck · A-201 · Railing plan + elevation

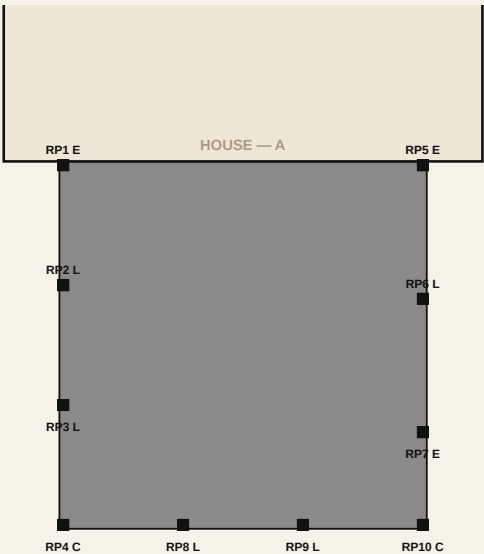
A-201

RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...

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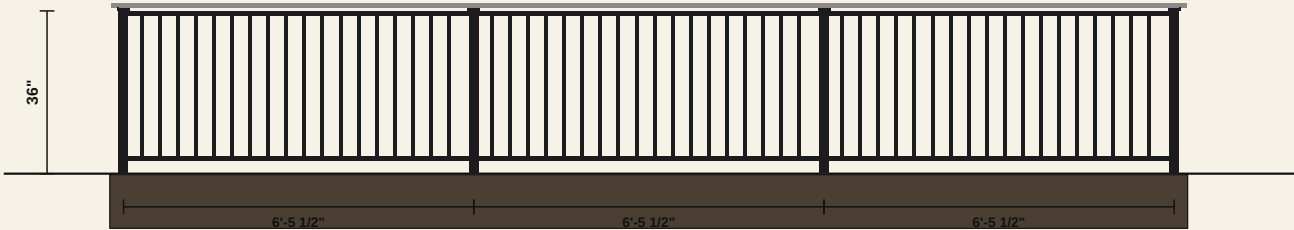
RAILING

Fulton 36" Black: 8 bays, 10 posts, 54 LF + drink rail

Fulton 36" — 10 posts (2 CORNER, 3 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn

Bays: left 6'-5 1/2" · left 6'-5 1/2" · left 6'-5 1/2" · right 7'-2 1/4" · right 7'-2 1/4" · front 6'-5 1/2" · front 6'-5 1/2" · front 6'-5 1/2"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.



Plan above · elevation of the outer edge below

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20 x 20 deck · A-301 · Section + front elevation

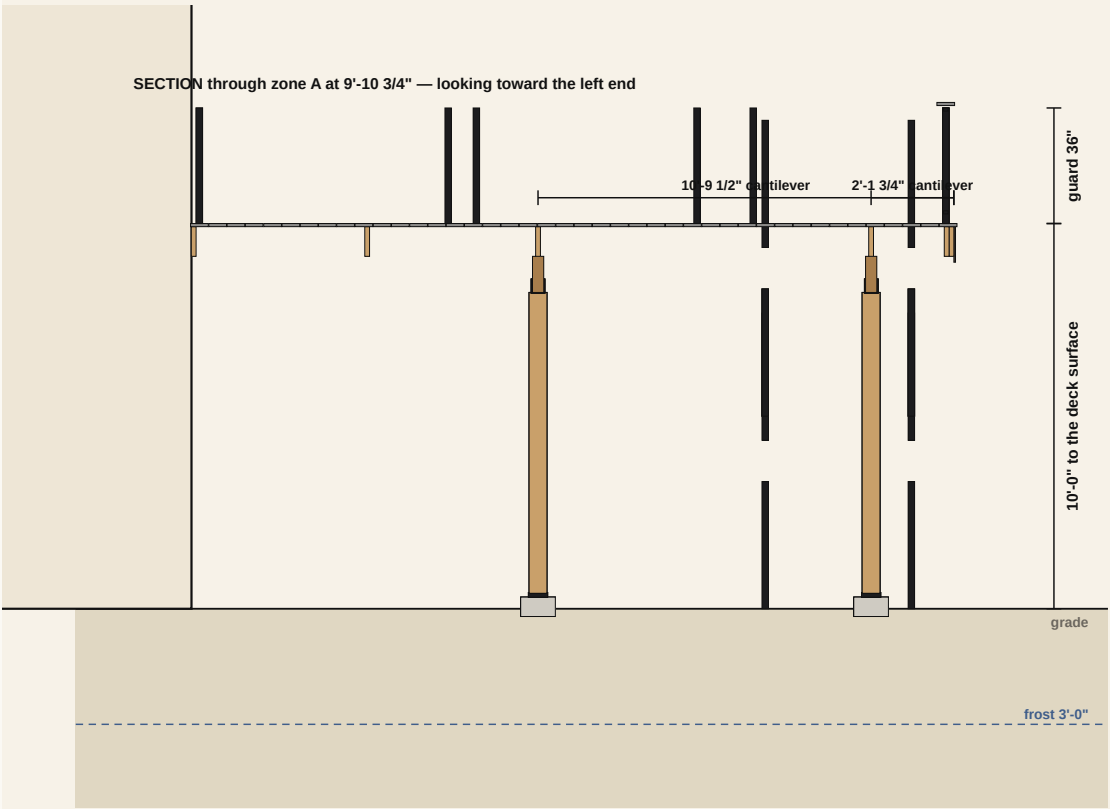
A-301

SECTION + FRONT ELEVATION

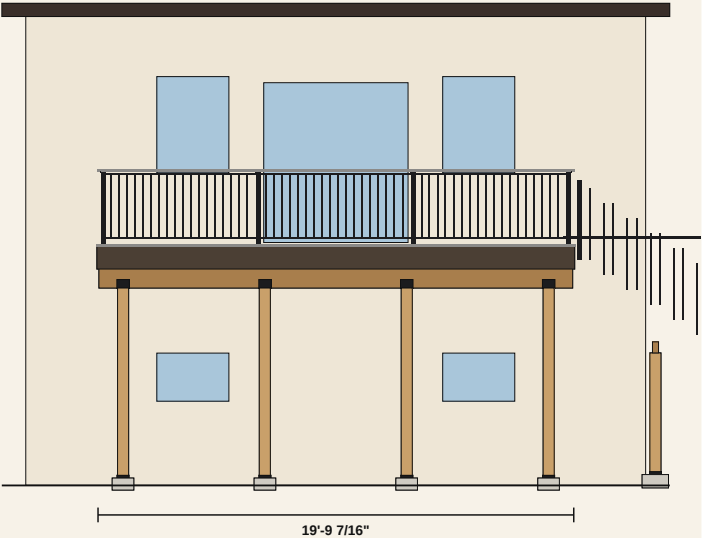
Cut through the deepest zone; heights to grade, frost line, cantilever, guard

20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...

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FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb); DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8"

Posts 12 x 6x6 on DP-50/50

2x10 ledger x 19'-9 7/16" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

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20 x 20 deck · D-501 · Details

D-501

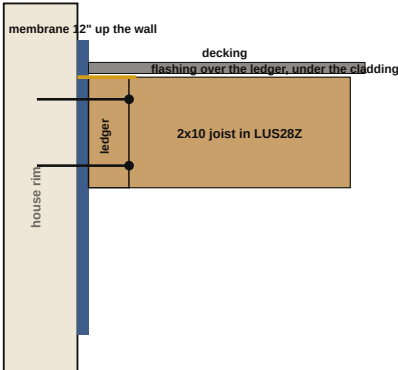
DETAILS

Ledger · post / beam / cap · footing · rail post · deck edge · the job's special condition

20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...

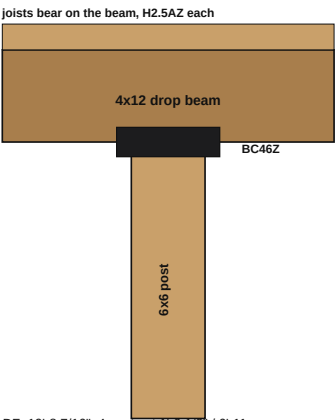
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1 LEDGER AT THE HOUSE



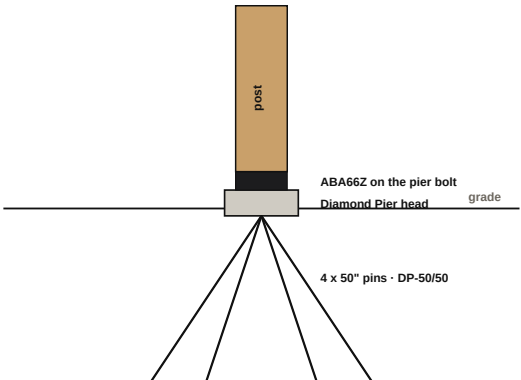
2x10 ledger x 19'-9 7/16" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners

2 POST · BEAM · CAP

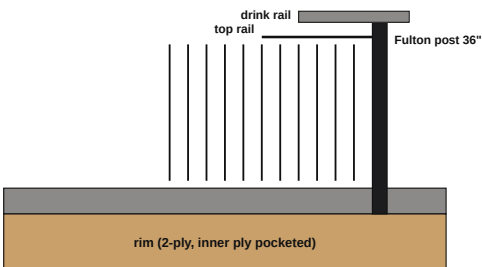


DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb); DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-

3 FOOTING · POST BASE

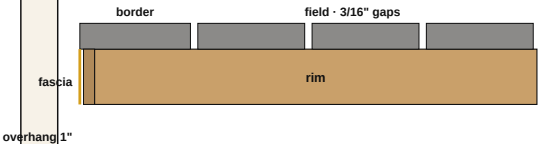


4 RAIL POST · GUARD



Fulton 36" — 10 posts (2 CORNER, 3 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn

5 DECK EDGE · BORDER



Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1726 field + 110 border/divider)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.

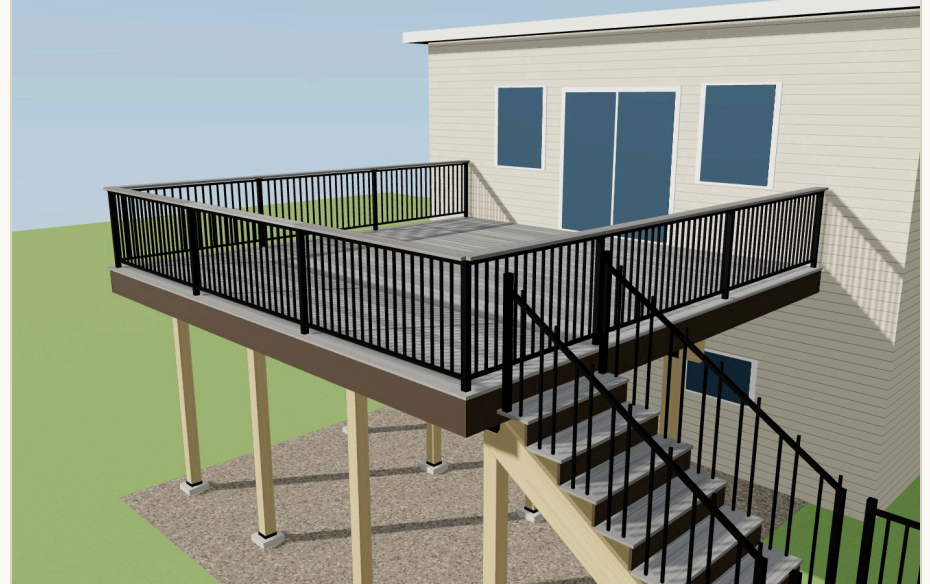
Schematic details, not to scale

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20 x 20 deck



From the yard



From the front corner



Isometric



Under the deck

